

✓ Optical:

$$\textcircled{a} \quad L_{\text{edd}} = a \frac{M_{\text{BH}}}{M_{\odot}}$$

$$M(t) = M_0 e^{kt} \quad \left| \quad k = \frac{1}{t_{\text{edd}}} = \frac{4\pi G M_H}{E \sigma_T c}$$

For $E = 0.1$:

$$k \approx (4.5 \times 10^7 \text{ yr})^{-1}$$

$$M(t_2) = 2 M(t_1)$$

$$e^{t_2 k} = 2 e^{t_1 k}$$

$$e^{k(t_2 - t_1)} = 2$$

$$\ln(e^{(t_2 - t_1)k}) = \ln(2)$$

$$(t_2 - t_1)/t_{\text{edd}} = \ln(2)$$

$$\Delta t = \ln(2) t_{\text{edd}} = \ln(2) 4.5 \times 10^7 \text{ yrs} \\ \approx 3.12 \times 10^7 \text{ yrs}$$

$$\textcircled{b} \quad \underline{z = 7.1}, \quad t_{\text{un.}} \approx 0.77 \text{ Gyr}$$

$$L_{\text{obs}} = 6.3 \times 10^{13} L_{\odot} = 1.26 \times 10^{31} \left(\frac{M_{\text{BH}}}{M_{\odot}} \right) W$$

$$M_{\text{now}} = M_0 e^{kt_{\text{un.}}}$$

